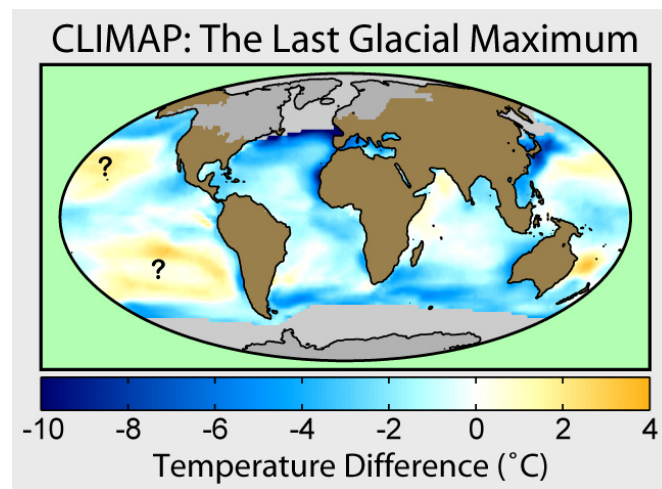


Climate and Ecology of the Pleistocene



Climate and Ecology of the Pleistocene

The climate of the Pleistocene was colder and drier than today in most parts of the world; due to this, glaciers covered what is today Canada, the northeastern US, northern Europe (Scandinavia and the British Isles), and Siberia. Glaciers that exist today were much larger, such as those of the Rockies, the Alps, the Andes, the Himalayas, Greenland, and Antarctica. Few forests existed, mainly in equatorial regions, and grasslands and tundra dominated the Northern Hemisphere. Many warm-blooded animals adapted to these conditions by increasing their size and their volume-to-surface area ratio to limit the loss of heat. These **megafauna** went extinct in the later Pleistocene, often coincident with the arrival of humans. Of course, smaller animals, such as rodents and birds, were also common, and those species with better abilities to evade human predators persisted into recent periods. Remember that the Pleistocene was also an era of several climatic fluctuations, some being extremely rapid, by perhaps as much as 10°C over a single decade. Compare that with the relatively (for the Holocene) “rapid” pace of climate change today, with global temperatures having risen by about 1°C (<https://www.climate.gov/news-features/understanding-climate/climate-change-global-temperature>) between 1970 and 2020. As we have trouble adapting to contemporary climate change, imagine those living in the Pleistocene; their grandparents’ world was one of ice, and theirs, perhaps, was one markedly warmer. It has been argued that this instability—rather than the cold, dry average climate—prevented humans from developing agriculture during the Pleistocene.

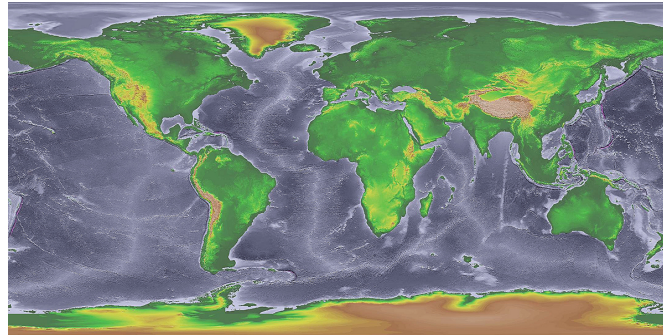


Global temperatures during the Last Glacial Maximum, c. 25,000 years ago, as differences from 20th-century CE averages, in degrees Celsius. Note that the coldest temperatures, in the North Atlantic and Pacific, are around 10°C

(or 18°F) colder, on average, than those of the 20th century, which are several degrees cooler than averages today. Map produced by CLIMAP computer model.

Source: [Wikimedia Commons](#)

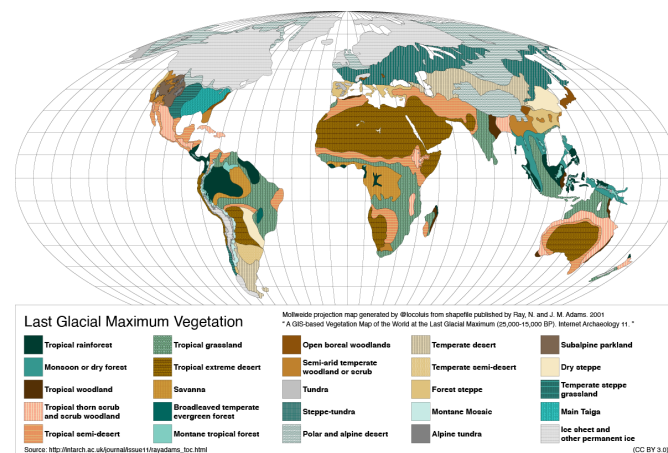
(<https://commons.wikimedia.org/wiki/File:CLIMAP.jpg>)



Map of sea levels during the Last Glacial Maximum, c. 25,000 years ago. This represents a sea level around 120 meters below that of the present. Note that the North Sea and Baltic Sea do not exist; the Black Sea and Red Sea are lakes; much of Indonesia is a solid land mass, as is Australia/Tasmania/New Guinea; and Japan is connected to the Asian mainland. Most importantly, Asia and North America are connected across what is now the Bering Strait.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Global_sea_levels_during_the_last_Ice_Age.jpg)



Map of vegetation communities during the Last Glacial Maximum. Observe the large areas of glacial ice covering portions of the Northern Hemisphere, and the large areas of steppe-

tundra and polar desert that cover much of the remainder of North America and Eurasia. Forests were extremely limited in extent due to the cold, dry climate that dominated globally.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Last_Glacial_Maximum_Vegetation_Map.svg)



A late Pleistocene landscape of northern Spain, as illustrated by Mauricio Antón. This painting depicts horses, woolly mammoths (*Mammuthus primigenius*), European cave lions (*Panthera leo spelaea*) with a reindeer carcass, and a woolly rhinoceros (*Coelodonta antiquitatis*). Mammoths, cave lions, and a woolly rhinoceros are **megafauna** that went extinct by the end of the Pleistocene.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Ice_age_fauna_of_northern_Spain_-_Mauricio_Ant%C3%B3n.jpg)

Pleistocene megafauna are possibly the coolest animals in the history of the planet! Imagine armadillos the size of a hippo ([glyptodonts](https://en.wikipedia.org/wiki/Glyptodont) (<https://en.wikipedia.org/wiki/Glyptodont>)), a [bear](https://en.wikipedia.org/wiki/Tremarctinae) (<https://en.wikipedia.org/wiki/Tremarctinae>) weighing up to 1,000 kilograms and standing 2.5–3.0 meters (8–10 feet!) tall, and [sloths](https://en.wikipedia.org/wiki/Ground_sloth) (https://en.wikipedia.org/wiki/Ground_sloth) larger than an African elephant! Camels and horses roamed the plains of North America, while a 1-meter (3-foot)-tall

flightless owl

(<https://en.wikipedia.org/wiki/Ornimegalonyx>) was an ambush predator of poisonous rats in Cuba. Nearly all of these creatures went extinct during the late Pleistocene or early Holocene; some (such as the elephant, rhinoceros, and hippo) still exist, mainly in Africa. It is still hotly debated whether human hunting or climate change was the primary reason for their extinction, though it is highly likely that human hunting played at least some role in this process, based on the coincidence of human arrivals and widespread extinctions around the world during different climatic periods. Feel free to learn more on Wikipedia

(https://en.wikipedia.org/wiki/Late_Pleistocene_extinctions) or at these links on the extinction of megafauna in Australia (<https://www.unsw.edu.au/newsroom/news/2017/01/climate-change-helped-kill-off-super-sized-ice-age-animals-in-au>), Beringia (<https://www.nps.gov/articles/aps-17-1-4.htm>), or southern California (<https://tarpits.org/research-collections>)!